PoLoRA as steepest descent under a gradient metric

direction by a metric LMO, magnitude by a trust region

Derivation draft — for discussion, not for `main.tex`.

1 Setup

A LoRA layer adds the rank-r adapter $W = BA \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ to a frozen weight ($A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$). A step $\Delta A, \Delta B$ changes W to first order by the tangent $U := B\Delta A + \Delta B A$ (dropping $\Delta B \, \Delta A$). With $G = \nabla_W \mathcal{L}$ and factor gradients $G_A := B^\top G$, $G_B := G A^\top$, the loss decrease splits,

$$\langle G, U \rangle = \langle G_A, \Delta A \rangle + \langle G_B, \Delta B \rangle. \tag{1}$$

$\langle X, Y \rangle = \operatorname{tr}(X^\top Y)$	Frobenius inner product
$\lVert\cdot\rVert_2,\ \lVert\cdot\rVert_*,\ \lVert\cdot\rVert_F$	spectral, nuclear (dual of spectral), Frobenius norms
$\operatorname{polar}(M) = UV^\top$	for $M = U\Sigma V^\top$ with singular values set to 1
$P, Q \succ 0$	symmetric, Kronecker ansatz $\Sigma = Q \otimes P$ for the gradient 2nd moment (§6)

In the optimizer G_A, G_B are momentum buffers. The derivation uses raw gradients.

2 The metric

A sample gives a rank-one pseudo-gradient $G_i = b_i x_i^\top$ (x_i input, b_i output sensitivity), disturbing the loss to first order by $\langle G_i, U \rangle = b_i^\top U x_i$. The worst single-sample disturbance, $\max_i |\langle G_i, U \rangle|$, is not enumerable. PoLoRA surrogates it by the worst over rank-one pseudo-gradients of unit metric size.

Lemma 1 (Metric Surrogate). *Under the ansatz $\Sigma = Q \otimes P$, with $\lVert R \rVert_{\Sigma^{-1}}^2 := \operatorname{vec}(R)^\top \Sigma^{-1} \operatorname{vec}(R)$,*

$$\Gamma(U) := \sup_{\lVert bx^\top \rVert_{\Sigma^{-1}} \le 1} |b^\top U x| = \lVert P^{1/2} U Q^{1/2} \rVert_2. \tag{2}$$

Proof. The ansatz factors the size, $\lVert bx^\top \rVert_{\Sigma^{-1}}^2 = (b^\top P^{-1} b)(x^\top Q^{-1} x)$. Substituting $b = P^{1/2} a$, $x = Q^{1/2} z$ makes the objective $a^\top (P^{1/2} U Q^{1/2}) z$, with supremum $\lVert P^{1/2} U Q^{1/2} \rVert_2$ over $\lVert a \rVert \lVert z \rVert \le 1$. □

The metric spectral norm Γ is thus a surrogate for the worst-sample disturbance, not the disturbance itself. (Σ^{-1} matches the pseudo-gradient set's shape, Appendix A. The mean-square response gives the Frobenius surrogate, Appendix B.)

3 The step: ideal program and surrogate

PoLoRA is steepest descent in the metric Γ, split (as in Muon, Scion) into a direction and a magnitude:

$$\begin{aligned} \text{direction:} \quad & \operatorname*{arg\,min}_{\Gamma(U)\le 1} \langle G, U\rangle, \\ \text{magnitude:} \quad & \|U\|_2 \le \eta. \end{aligned} \tag{3}$$

The LMO is scale-invariant, so the direction step fixes direction only. The trust region sets the magnitude. The objective (1) is separable across factors, but Γ is on the joint tangent $U = B\Delta A + \Delta B A$ (Lemma 1), so the ideal direction couples ΔA and ΔB. Section 4 surrogates it, and Section 5 sets the magnitude.

4 Stage one: the direction

By subadditivity of the spectral norm,

$$\|P^{1/2}(B\Delta A + \Delta B A)Q^{1/2}\|_2 \le \|P^{1/2}B\Delta A\, Q^{1/2}\|_2 + \|P^{1/2}\Delta B\, A\, Q^{1/2}\|_2, \tag{4}$$

so bounding the two single-factor responses bounds Γ. PoLoRA surrogates the joint constraint by these two, descending each factor in its induced metric. Each whitens to an $r \times r$ form,

$$\|P^{1/2}B\Delta A\, Q^{1/2}\|_2 = \|C_A^{1/2}\Delta A\, Q^{1/2}\|_2, \qquad \|P^{1/2}\Delta B\, A\, Q^{1/2}\|_2 = \|P^{1/2}\Delta B\, C_B^{1/2}\|_2, \tag{5}$$

with $C_A := B^\top P B$, $C_B := AQA^\top$ (via $P^{1/2}B = VC_A^{1/2}$, $V^\top V = I$, orthogonal V preserving the spectral norm). The matrices C_A, C_B are the factor-gradient covariances (§6).

Lemma 2 (Whitened-Polar LMO)**.** *For symmetric positive definite L, R and any M, with $H := L^{-1}MR^{-1}$,*

$$\operatorname*{arg\,min}_{\|LXR\|_2 \le 1} \langle M, X\rangle = -L^{-1}\operatorname{polar}(H)R^{-1}. \tag{6}$$

Proof. $Y = LXR$ gives $\langle M, X\rangle = \langle H, Y\rangle$ with $\|Y\|_2 \le 1$. By spectral–nuclear duality the minimum is $-\|H\|_*$, attained at $Y = -\operatorname{polar}(H)$. □

Proposition 3 (Per-Factor Directions)**.** *The objective* (1) *under the two per-factor constraints separates into two LMOs, solved by*

$$W_A = -C_A^{-1/2}\operatorname{polar}(H_A)Q^{-1/2}, \qquad W_B = -P^{-1/2}\operatorname{polar}(H_B)C_B^{-1/2}, \tag{7}$$

with $H_A := C_A^{-1/2}G_A Q^{-1/2}$ and $H_B := P^{-1/2}G_B C_B^{-1/2}$.

Proof. Apply Lemma 2 with $(L, R, M) = (C_A^{1/2}, Q^{1/2}, G_A)$ and $(P^{1/2}, C_B^{1/2}, G_B)$. □

The oracle is scale-invariant. Keep the unit directions

$$\widehat{W}_A := W_A/\|W_A\|_2, \qquad \widehat{W}_B := W_B/\|W_B\|_2.$$

5 Stage two: the magnitude

The directions $\widehat{W}_A, \widehat{W}_B$ need a magnitude. A fixed one does not give a useful step.

Lemma 4 (Gauge-Unbounded Factor Step)**.** *For a fixed ρ, the factor step $\Delta A = \rho\widehat{W}_A$, $\Delta B = \rho\widehat{W}_B$ leaves the weight change unbounded over the factor gauge $(B, A) \mapsto (Bc, A/c)$,*

$$\sup_{c>0} \|U\|_2 = \infty. \tag{8}$$

Proof. The gauge fixes $W = BA$ and (via H_A, H_B invariant) the directions $\widehat{W}_A, \widehat{W}_B$, so $U = \rho(c\,B\widehat{W}_A + c^{-1}\widehat{W}_B A)$ and $\|U\|_2 \to \infty$ as $c \to 0$ and as $c \to \infty$. □

PoLoRA budgets the weight change directly instead.

Proposition 5 (Step Magnitude)**.** *The trust region $\|U\|_2 \leq \eta$ with the symmetric tie-break $\|\Delta A\|_2 = \|\Delta B\|_2 = \rho$ gives the step*

$$\Delta A = \rho\,\widehat{W}_A, \quad \Delta B = \rho\,\widehat{W}_B, \quad \rho = \frac{\eta}{\|A\|_2 + \|B\|_2}, \tag{9}$$

which holds $\|U\|_2 \leq \eta$ in every gauge.

Proof. The product bound gives $\|U\|_2 \leq \|B\|_2\|\Delta A\|_2 + \|A\|_2\|\Delta B\|_2 = (\|A\|_2 + \|B\|_2)\rho$. Setting this to η gives ρ. In the gauge of Lemma 4 $\rho = \eta/(c\|B\|_2 + c^{-1}\|A\|_2)$, and the bound stays $\leq \eta$. □

A fixed factor step fails because the weight-change size $\|U\|_2$ is blind to part of the step. It is zero for a whole family of nonzero factor moves, $\Delta A = KA$ and $\Delta B = -BK$ with $K \in \mathbb{R}^{r\times r}$, since $B(KA) - (BK)A = 0$, so its ball runs off to infinity along them and cannot pick out a direction. The metric direction norm has no such blind spot, being zero only at $\Delta A = \Delta B = 0$. So the metric picks the direction and the weight-change budget only sets its magnitude.

6 Estimating the metric

P, Q follow from two moment identities of the ansatz, with expectations over per-example gradients and any compatible M, N:

$$\mathbb{E}[GMG^\top] = \mathrm{tr}(MQ)\,P, \qquad \mathbb{E}[G^\top NG] = \mathrm{tr}(NP)\,Q. \tag{10}$$

Lemma 6 (Moment Closure)**.** *With $C_A = B^\top PB$ and $C_B = AQA^\top$,*

$$\mathbb{E}\left[\tfrac{1}{r}G_A^\top C_A^{-1} G_A\right] = Q, \qquad \mathbb{E}\left[\tfrac{1}{r}G_B C_B^{-1} G_B^\top\right] = P. \tag{11}$$

Proof. Apply the second identity of (10) with $N = BC_A^{-1}B^\top$. Then $G^\top NG = G_A^\top C_A^{-1} G_A$ and $\mathrm{tr}(NP) = \mathrm{tr}(I_r) = r$. The other uses $M = A^\top C_B^{-1} A$. □

PoLoRA keeps the diagonals by EMA:

$$q_j \leftarrow \mathrm{EMA}\left[\tfrac{1}{r}(G_A)_{:,j}^\top C_A^{-1}(G_A)_{:,j}\right], \qquad p_i \leftarrow \mathrm{EMA}\left[\tfrac{1}{r}(G_B)_{i,:} C_B^{-1}(G_B)_{i,:}^\top\right]. \tag{12}$$

Because C_A depends on P and C_B on Q, the optimizer takes one EMA alternation per step, then freezes P, Q for the step. The inverse square roots are damped.

A The metric is the inverse second moment

Let $g = \mathrm{vec}(G_i)$ and $\Sigma = \mathbb{E}[gg^\top] \succ 0$. The set's extent along a unit direction u is the root-mean-square component $s(u) = \sqrt{u^\top \Sigma u}$, built from Σ alone and (by polarization) the full second-order shape.

Lemma 7 (Second-Moment Ellipsoid)**.** *Among centered ellipsoids* $E_A = \{g : g^\top A g \leq 1\}$*,* $A \succ 0$*, the unique one whose support width matches the extent in every direction,* $\max_{g \in E_A} u^\top g = s(u)$ *for all* u*, has* $A = \Sigma^{-1}$*.*

Proof. The support width of E_A is $\sqrt{u^\top A^{-1} u}$. Equating to $s(u) = \sqrt{u^\top \Sigma u}$ for all u gives $A^{-1} = \Sigma$. □

So the unit ball of $\|\cdot\|_{\Sigma^{-1}}$ reaches, in each direction, as far as the pseudo-gradients spread.

B Mean-square response: the Frobenius surrogate

Lemma 8 (Mean-Square Response)**.** *Under the ansatz* $\Sigma = Q \otimes P$*, the average squared sample interaction is*

$$\mathbb{E}_i[\langle G_i, U \rangle^2] = \mathrm{vec}(U)^\top \Sigma \ \mathrm{vec}(U) = \|P^{1/2} U Q^{1/2}\|_F^2. \tag{13}$$

Its LMO replaces the spectral–nuclear duality of Lemma 2 by Cauchy–Schwarz ($\mathrm{polar}(H) \to H/\|H\|_F$), giving the per-factor directions $W_A^{\mathrm{F}} \propto C_A^{-1} G_A Q^{-1}$ and $W_B^{\mathrm{F}} \propto P^{-1} G_B C_B^{-1}$. Worst-case over samples surrogates the disturbance by the spectral polar direction. Mean-square surrogates it by the Frobenius inverse direction.